

Concentration Bounds: Proofs and Examples

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1 Markov's Inequality

Theorem: Let X be a non-negative random variable ($X \geq 0$) with expected value $E[X]$, and let $a > 0$. Then:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Proof (for continuous variables): Let $f(x)$ be the probability density function of X . The expected value is defined as:

$$E[X] = \int_0^{\infty} x f(x) dx$$

We can split this integral at the threshold a :

$$E[X] = \int_0^a x f(x) dx + \int_a^{\infty} x f(x) dx$$

Since X is non-negative and $f(x) \geq 0$, the first integral is non-negative. Dropping it yields an inequality:

$$E[X] \geq \int_a^{\infty} x f(x) dx$$

In the interval $[a, \infty)$, the minimum value x can take is a . Substituting x with a makes the right side even smaller:

$$E[X] \geq \int_a^{\infty} a f(x) dx$$

Factoring out the constant a :

$$E[X] \geq a \int_a^{\infty} f(x) dx$$

By definition, the remaining integral is the probability that $X \geq a$:

$$E[X] \geq a P(X \geq a)$$

Dividing by a (since $a > 0$) completes the proof:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Example: Consider monitoring the response time T (in seconds) of a Large Language Model in production. We know $T \geq 0$ and historical data shows the expected response time is $E[T] = 2$ seconds. We want to find the maximum probability that a specific query takes 10 seconds or more ($a = 10$).

$$P(T \geq 10) \leq \frac{2}{10} = 0.2$$

Thus, there is at most a 20% chance that the model takes 10 seconds or longer to respond.

2 Chebyshev's Inequality

Theorem: Let X be a random variable with finite expected value μ and finite non-zero variance σ^2 . For any real number $a > 0$:

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Proof: We define a new random variable $Y = (X - \mu)^2$. Since the square of any real number is non-negative, $Y \geq 0$. This allows us to apply Markov's Inequality to Y using a threshold a^2 :

$$P(Y \geq a^2) \leq \frac{E[Y]}{a^2}$$

Substitute Y back with $(X - \mu)^2$. Notice that $E[(X - \mu)^2]$ is the mathematical definition of the variance σ^2 :

$$P((X - \mu)^2 \geq a^2) \leq \frac{\sigma^2}{a^2}$$

The event $(X - \mu)^2 \geq a^2$ is algebraically identical to $|X - \mu| \geq a$. Taking the square root of both sides inside the probability yields:

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Example: Suppose an RL (Reinforcement Learning) agent is simulating intervention strategies for a virus. Let R be the accumulated reward, with an expected mean of $\mu = 500$ and a variance of $\sigma^2 = 400$. We want to find the probability that the agent's performance in a given run deviates by 60 points or more from its expected mean ($a = 60$).

$$P(|R - 500| \geq 60) \leq \frac{400}{60^2} = \frac{400}{3600} = \frac{1}{9} \approx 0.111$$

The probability of the agent's reward deviating by 60 points or more is at most 11.1%.

3 Hoeffding's Inequality

Theorem: Let $X_1, X_2, \dots, X_n$ be independent random variables strictly bounded such that $a_i \leq X_i \leq b_i$ almost surely. Let $S_n = \sum_{i=1}^n X_i$. For any $t > 0$:

$$P(|S_n - E[S_n]| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$$

Proof: First, center the variables by defining $Y_i = X_i - E[X_i]$. Consequently, $E[Y_i] = 0$ and $Y_i \in [a_i - E[X_i], b_i - E[X_i]]$, meaning the interval length remains $b_i - a_i$. We evaluate the one-sided probability $P(\sum_{i=1}^n Y_i \geq t)$. Using Chernoff's method, for any $s > 0$:

$$P\left(\sum_{i=1}^n Y_i \geq t\right) = P\left(\exp\left(s \sum_{i=1}^n Y_i\right) \geq \exp(st)\right)$$

Applying Markov's inequality to the right side:

$$P\left(\sum_{i=1}^n Y_i \geq t\right) \leq e^{-st} E\left[\exp\left(s \sum_{i=1}^n Y_i\right)\right]$$

Because the variables are independent, the expected value of their product is the product of their expected values:

$$P\left(\sum_{i=1}^n Y_i \geq t\right) \leq e^{-st} \prod_{i=1}^n E[\exp(sY_i)]$$

Assuming **Hoeffding's Lemma**, since $E[Y_i] = 0$ and Y_i is bounded within an interval of length $(b_i - a_i)$, we have $E[\exp(sY_i)] \leq \exp\left(\frac{s^2(b_i - a_i)^2}{8}\right)$. Substituting this:

$$P\left(\sum_{i=1}^n Y_i \geq t\right) \leq \exp\left(-st + \frac{s^2}{8} \sum_{i=1}^n (b_i - a_i)^2\right)$$

To get the tightest bound, we minimize the exponent with respect to s . Setting the derivative to zero yields $s = \frac{4t}{\sum_{i=1}^n (b_i - a_i)^2}$. Substituting this optimal s back into the inequality gives the one-sided bound:

$$P(S_n - E[S_n] \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$$

By symmetry, the probability of the opposite tail $P(S_n - E[S_n] \leq -t)$ yields the same bound. Using Boole's inequality (union bound) for the absolute value $|S_n - E[S_n]| \geq t$, we add both tails, introducing a factor of 2:

$$P(|S_n - E[S_n]| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$$

Example: Suppose an LLM is evaluated on a benchmark consisting of $n = 1000$ independent binary questions. Let $X_i = 1$ if the answer is correct and $X_i = 0$ if incorrect, meaning bounds are $[a_i, b_i] = [0, 1]$ and $(b_i - a_i)^2 = 1$ for all i . Let S_{1000} be the total empirical score. We want to bound the probability that the empirical score deviates from the true expected score $E[S_{1000}]$ by $t = 50$ points or more.

$$\begin{aligned} P(|S_{1000} - E[S_{1000}]| \geq 50) &\leq 2 \exp\left(-\frac{2(50)^2}{\sum_{i=1}^{1000} 1^2}\right) \\ &= 2 \exp\left(-\frac{2(2500)}{1000}\right) \\ &= 2 \exp(-5) \approx 2(0.0067) = 0.0134 \end{aligned}$$

There is at most a 1.34% chance that the benchmark score misrepresents the true expected performance by 50 points or more.